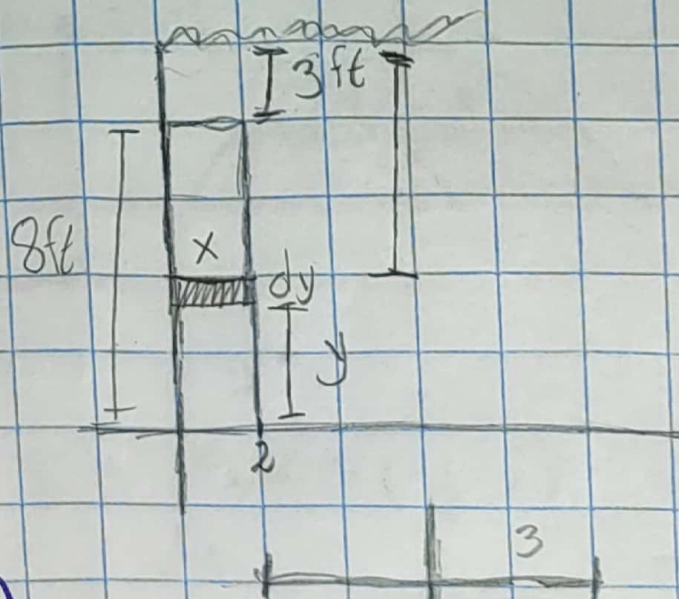


597

Tarea sección 8.3

3)



$$F = \rho g d A$$

$$\rho g = 62.4$$

$$d = 11 - y$$

$$A = 2 dy$$

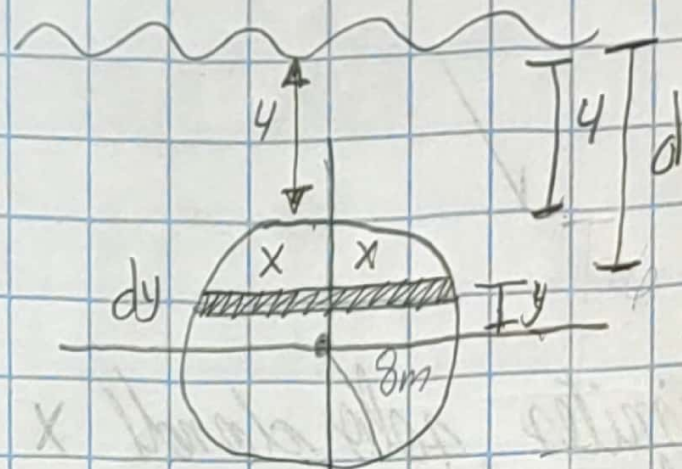
$$F = \int_0^8 62.4 (11 - y) 2 dy$$

$$F = 124.8 \int_0^8 11 - y dy$$

$$F = 124.8 \left[11y - \frac{1}{2}y^2 \right]_0^8$$

$$F = \underline{6988.8 \text{ lb}}$$

5



$$F = \rho g d A$$

$$\rho g = 9800 \text{ N/m}^3$$

$$d = 12 - y$$

$$A = 2x dy = \sqrt{64 - y^2} dy$$

$$F = \int_{-8}^8 9800 (12 - y) 2 (\sqrt{64 - y^2}) dy$$

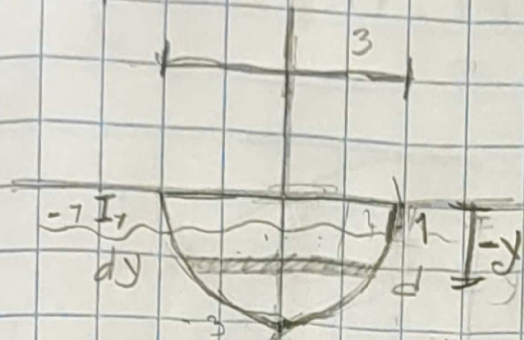
$$F = 23644882.95 \text{ N}$$

$$x^2 + y^2 = r^2$$

$$x^2 + y^2 = 8^2$$

$$x = \sqrt{64 - y^2}$$

7)



$$x^2 + y^2 = r^2$$

$$x^2 + y^2 = 9$$

$$x = \sqrt{9 - y^2}$$

$$-y = -1$$

$$-y + 1 = 2$$

$$F = \rho g d A$$

$$F = 9800 \text{ N/m}^3$$

$$d = -1 + y$$

$$A = 2x dy$$

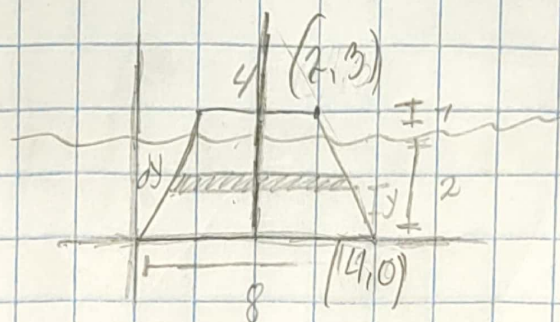
$$A = 2\sqrt{9 - y^2} dy$$

$$F = \int_{-3}^{-1} 9800(-1 - y) 2\sqrt{9 - y^2} dy$$

$$F = 19600 \int_{-3}^{-1} (-1 - y) \sqrt{9 - y^2} dy$$

$$F =$$

9)



$$F = \rho g d A$$

$$\rho g = 62.4$$

$$d = 2 - y$$

$$A = 2x dy$$

$$A = 2\left(\frac{2}{3}y - 4\right) dy$$

$$F = \int_0^2 62.4(2 - y) 2\left(\frac{2}{3}y - 4\right) dy$$

$$F = 124.8 \int_0^2 (2 - y) \left(\frac{2}{3}y - 4\right) dy$$

$$F = 887.47$$

$$m = \frac{3 - 0}{2 - 4} = -\frac{3}{2}$$

$$-\frac{2}{3}y - 4$$

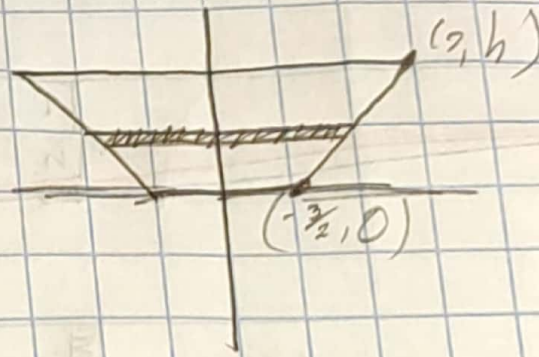
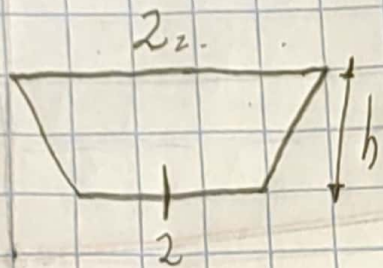
$$y - y_0 = m(x - x_0)$$

$$y - 0 = -\frac{3}{2}(x - 4)$$

$$y = -\frac{3}{2}x + 6$$

$$\frac{2}{3}y - 4 = x$$

11)



$$F = \rho g d A$$

$$A = 2 x dy = 2 \left(\frac{2y}{2h}, \frac{z}{2} \right) dy$$

$$F = \rho g \cdot 2 \int_0^h \left(h - y \right) \left(\frac{2y}{2} + \frac{2h}{2} \right) dy$$

$$F = \rho g \cdot 2 \int_0^h \left(\frac{2y}{2} - \frac{2h}{2} - \frac{2y^2}{2h} - \frac{2y}{2} \right) dy$$

$$F = \rho g \cdot 2 \int_0^h \left(\frac{2h}{2} - \frac{2y^2}{2h} \right) dy$$

$$F = \rho g \cdot 2 \left(\frac{2h^2}{2} - \frac{2h^2}{2} \right)$$

$$F = \rho g \left(2h^2 - \frac{2h^2}{3} \right) = \rho g \cdot \frac{2}{3} 2h^2$$

$$m = \frac{n}{1/28}$$

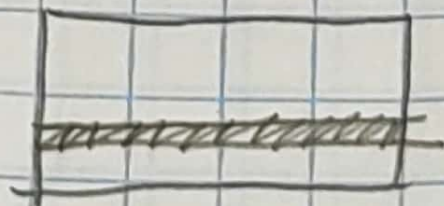
$$y = \left(\frac{n}{1/28} \left(x - \frac{9}{2} \right) \right)$$

$$y = \frac{2h}{2} x - h$$

$$x = \frac{2y}{2h} + \frac{2}{2}$$

17)

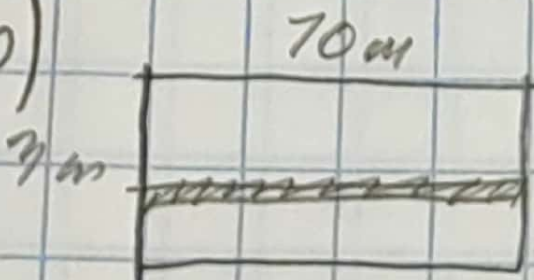
a)



$$F = 9800 \int_0^7 (7-y)(70) dy$$

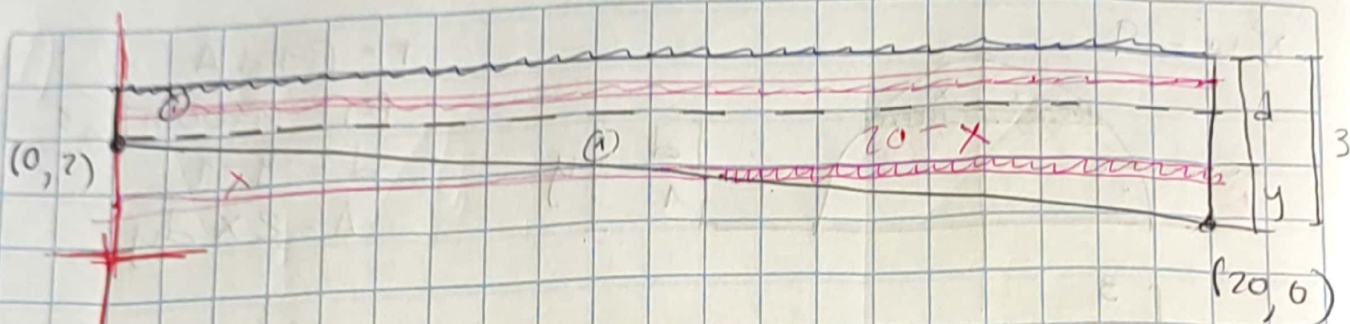
$$F = 49000 \text{ N}$$

b)



$$F = 9800 \int_0^3 (3-y)(70) dy$$

$$F = 447000 \text{ N}$$



$$\textcircled{1} F = \rho g d A$$

$$\rho g = 9800 \text{ N/m}^3$$

$$d = 3 - y$$

$$A = (20 - x) dy$$

$$A = [20 - 10(2 - y)] dy$$

$$m = \frac{0 - 2}{20 - 0} = -\frac{2}{20} = -\frac{1}{10}$$

$$y - 2 = -\frac{1}{10}x$$

$$-10(y - 2) = x$$

$$10(2 - y) = x$$

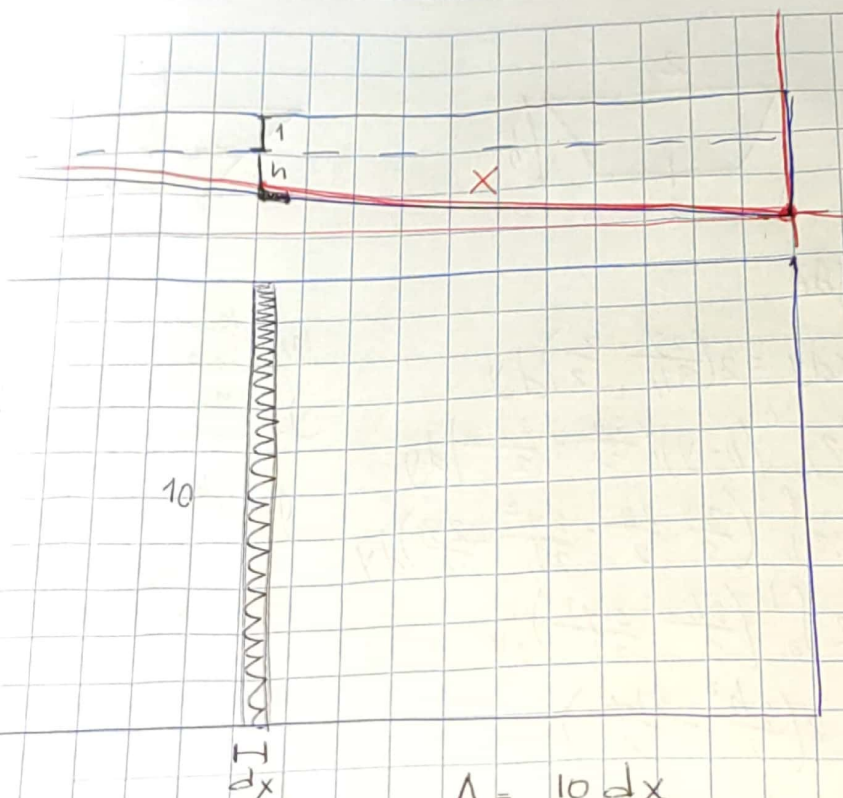
$$F = \int_0^2 9800(3 - y) [20 - 20 + 10y] dy$$

$$F = 9800 \int_0^2 (3 - y)(10y) dy = 98000 \int_0^2 (3 - y)y dy$$

$$F = 326,666.67 \text{ N}$$

$$\textcircled{2} F = \int_2^3 9800(3 - y) 20 dy = 98000$$

$$F = 424,666.67 \text{ N}$$



$$\rho g dA$$

$$\rho g = 9800$$

$$A = 10 dx$$

$$d = 1 + h$$

$$d = 1 + \frac{2(\sqrt{404} - x)}{\sqrt{404}}$$



$$\frac{2}{\sqrt{404}} = \frac{h}{\sqrt{404} - x}$$

$$\frac{2}{20} = \frac{h}{\sqrt{404}}$$

$$-x$$

$$1404 + x^2 = h^2$$

$$h = \frac{2(\sqrt{404} - x)}{\sqrt{404}}$$

$$F = 3939551.243$$

$$F = \int_0^{\sqrt{404}} 9800 \left[1 + \frac{2(\sqrt{404} - x)}{\sqrt{404}} \right] 10 dx$$

Значит